

Lecheng Ren

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Centre for Theoretical Physics, Queen Mary University of London, London E1 4NS, UK

Employment

Postdoctoral Researcher — Queen Mary University of London
2024 – 2027 — Royal Society Newton International Fellow (2025 – 2027)

Education

Bachelor's degree in Physics — Beijing Normal University
September 2015 – June 2019

Undergraduate Exchange Program in Physics — College of William and Mary
July 2017 – August 2017

Undergraduate research program — ITP, CAS
May 2018 – June 2019 — Advisor: Song He

PhD in Physics — Brown University
September 2019 – 2024 — Advisor: Anastasia Volovich

Publications

1. **L. Ren**, "de Sitter Wavefunction from Quadrangular Polylogarithms: Polygon Graphs," [arXiv:2610.06772], with L. Ferro, T. Lukowski, S. Paranjape, M. Spradlin, A. Volovich, H.-C. Weng, and Y.-Q. Zhang.
2. **L. Ren**, "Direct-to-Symbol Integration from Landau Analysis," [arXiv:2609.15952], with C. Larkin, A. J. McLeod, and A. Pokraka.
3. **L. Ren**, "Mass/electric versus NUT/magnetic charges: duality from scattering amplitudes in $D \geq 4$ and for all bosonic spins," [arXiv:2607.22938], with R. Monteiro and D. Siretanu.
4. **L. Ren**, "Energy-Energy Correlators at Strong Coupling," [arXiv:2607.21139], with M. Jackson, B. Wang, and C. Wen.
5. **L. Ren**, "A Graphical Coaction for FRW Integrals from Partial/Relative Twisted (Co)homology," [arXiv:2606.13627], with A. J. McLeod and A. Pokraka.
6. **L. Ren**, "de Sitter Wavefunction from Quadrangular Polylogarithms: Chain Graphs," [arXiv:2605.06542], with L. Ferro, T. Lukowski, M. Spradlin, A. Volovich, H.-C. Weng, and Y.-Q. Zhang.
7. **L. Ren**, "A Graphical Coaction for FRW Wavefunction Coefficients," [arXiv:2603.25703], with A. McLeod and A. Pokraka.
8. **L. Ren**, "Energy-energy correlator from the AdS Virasoro-Shapiro amplitude," Phys. Rev. D 113 (2026) 106013 [arXiv:2601.05312], with B. Wang and C. Wen.
9. **L. Ren**, "Superstring amplitudes from the field-theory limit: an n-point map at one loop," JHEP 01 (2026) 057 [arXiv:2510.02455], with R. Monteiro.
10. **L. Ren**, "A double copy from twisted (co)homology at genus g," [arXiv:2509.01598], with A. Pokraka and C. Rodriguez.

11. **L. Ren**, "Superstring amplitudes from BCJ numerators at one loop," JHEP 03 (2025) 017 [arXiv:2410.19663], with Y. Geyer, J. Guo, and R. Monteiro.

12. **L. Ren**, "Five-dimensional spinor helicity for all masses and spins," JHEP 07 (2025) 056 [arXiv:2405.09533], with A. Pokraka, S. Rajan, A. Volovich, and W. W. Zhao.

13. **L. Ren**, "A double copy from twisted (co)homology at genus one," JHEP 07 (2024) 040 [arXiv:2312.02148], with R. Bhardwaj, A. Pokraka, and C. Rodriguez.

14. **L. Ren**, "One-loop Integrals from Volumes of Orthoschemes," JHEP 05 (2024) 104 [arXiv:2306.04630], with M. Spradlin, C. Vergu, and A. Volovich.

15. **L. Ren**, "All-order celestial OPE from on-shell recursion," JHEP 10 (2023) 080 [arXiv:2305.11851], with A. Schreiber, A. Sharma, and D. Wang.

16. **L. Ren**, "Loop-level gluon OPEs in celestial holography," JHEP 11 (2022) 171 [arXiv:2208.14416], with R. Bhardwaj, L. Lippstreu, M. Spradlin, A. Y. Srikant and A. Volovich.

17. **L. Ren**, "On effective field theories with celestial duals," JHEP 08 (2022) 251 [arXiv:2206.08322], with M. Spradlin, A. Y. Srikant, and A. Volovich.

18. **L. Ren**, "Deformed $w_{1+\infty}$ Algebras in the Celestial CFT," SIGMA 19 (2023) 044 [arXiv:2111.11356], with J. Mago, A. Y. Srikant, and A. Volovich.

19. **L. Ren**, "Celestial dual superconformal symmetry, MHV amplitudes and differential equations," JHEP 12 (2021) 171 [arXiv:2106.16111], with Y. Hu, A. Y. Srikant, and A. Volovich.

20. **L. Ren**, "Symbol alphabets from tensor diagrams," JHEP 12 (2021) 079 [arXiv:2106.01405], with M. Spradlin and A. Volovich.

21. **L. Ren**, "Notes on polytopes, amplitudes and boundary configurations for Grassmannian string integrals," JHEP 04 (2020) 140 [arXiv:2001.09603], with S. He and Y. Zhang.

Presentations

1. Conference talk, "Energy-Energy Correlators at Strong Coupling," Pollica, September 2026.

2. Seminar talk, "Energy-Energy Correlators at Strong Coupling," Prague Spring Amplitudes 2, May 2026.

3. Conference talk, "Superstring amplitudes from the field-theory limit: an n-point map at one loop," YOUNGST@RS - Loops & Geometry: Hidden Structures in Multi-loop Amplitudes, Mainz Institute for Theoretical Physics, April 2026.

4. Invited talk, "Energy-Energy Correlators at Strong Coupling," Brown University, April 2026.

5. Conference talk, "Diagrammatic Coaction of Cosmological Correlators," The Mathematics behind Scattering Amplitudes, Galileo Galilei Institute, August 2024.

6. Conference talk, "Symbol Alphabets from Tensor Diagrams," Amplituhedra, Cluster Algebras, and Positive Geometry, Harvard CMSA, May 2024.

7. Invited talk, "The one-loop six-dimensional hexagon integral," Amplitudes Seminar, Bonn University, November 2023.

8. Gong show, "One-loop Integrals from Volumes of Orthoschemes," Amplitudes School 2023, CERN, August 2023.

9. Poster Presentation, "The twisted cohomology of Riemann-Wirtinger integrals," Amplitudes 2023, CERN, August 2023.

10. "On structures of celestial OPE and algebras," Amplifying Gravity at All Scales, Nordita, July 2023.

11. Invited talk, "On Effective field theory with celestial dual," Tianjin University, June 2023.

12. Invited talk, "On Effective field theory with celestial dual," Beijing Normal University, June 2023.

13. Invited talk, "On Effective field theory with celestial dual," ITP CAS, June 2023.
14. Invited talk, "One-loop Integrals from Volumes of Orthoschemes," ITP CAS, June 2023.
15. Poster Presentation, "Deformed $w_{1+\infty}$ Algebras in the Celestial CFT," Amplitudes 2022, Charles University, August 2022.
16. Poster Presentation, "Deformed $w_{1+\infty}$ Algebras in the Celestial CFT," String 2022, University of Vienna, July 2022.
17. Gong Show, "All-order OPE expansion from MHV amplitudes," Joint BHI/CMSA Conference on Flat Holography, CMSA, Harvard University, June 2022.
18. "On Effective field theory with celestial dual," Possible and Impossible in Effective Field Theory: From the S-Matrix to the Swampland, IAS, May 2022.
19. Gong Show, "Deformed $w_{1+\infty}$ Algebras in the Celestial CFT," Princeton Center for Theoretical Science, Princeton University, February 2022.
20. Short Talk, "Symbol Alphabets from Tensor Diagrams," Mainz Institute for Theoretical Physics, June 2021.

Fellowship and Awards

Newton International Fellowships — **The Royal Society**
Department of Physics and Astronomy, Queen Mary University of London — 2025 - 2027

Galkin Foundation Fellowship
Department of Physics, Brown University — 2023 - 2024

Joukowsky Presidential Fellowships
Department of Physics, Brown University — 2023 - 2024